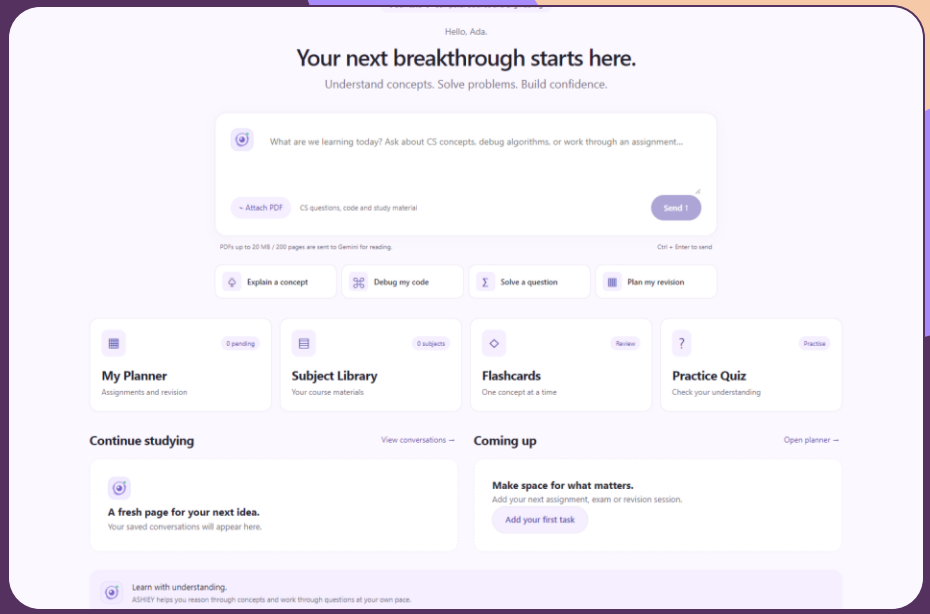
MID-TERM PROJECT DISPLAY

# StudyForge AI

A focused study companion that turns learning material into organised practice.




**Akshita · Aasveer Singh · Manik Jasrotia**

B.Tech project presentation

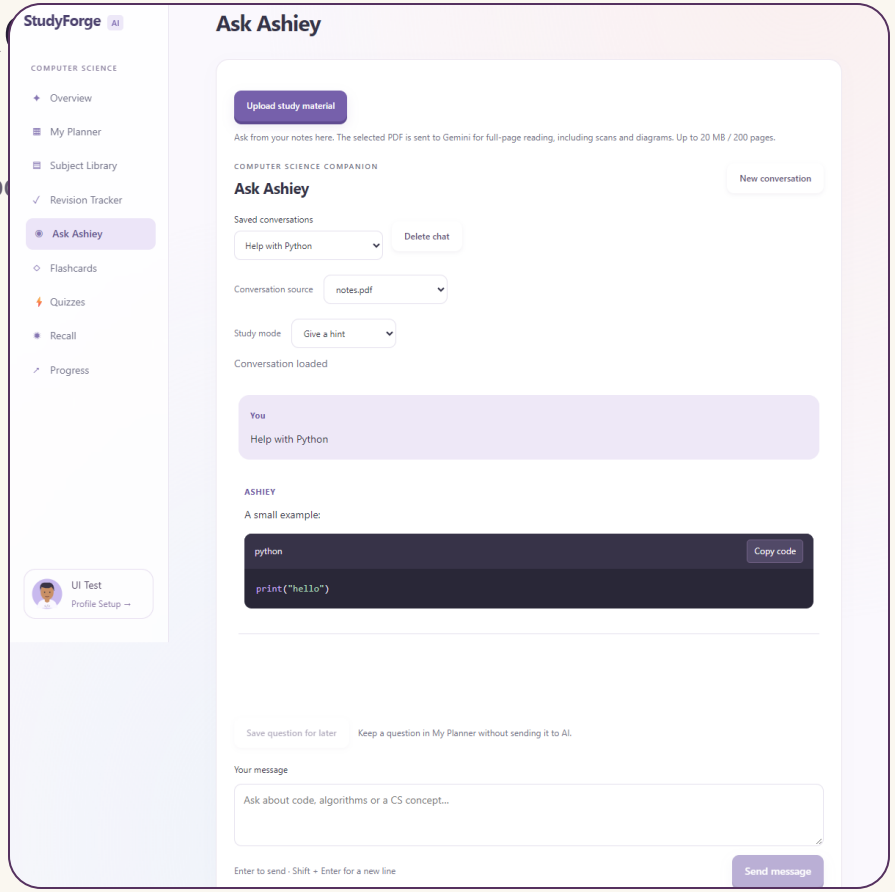
THE NEED

# Students need one calm place to learn, pra and ask.

Study material is commonly scattered across PDFs, notes, search tabs and separate quiz to

- Revision becomes slower when key points are hidden inside long notes.
- Generic practice questions repeat ideas and do not reflect the uploaded material.
- Students also need a simple way to track tasks, chat about a topic and return to their work.

DESIGNED FOR B.TECH STUDENTS



PROJECT OBJECTIVE

# Make study actions feel connected.

StudyForge brings material, AI assistance and personal organisation into one web application.

01

### Collect

Upload PDFs and keep study material available in a personal workspace.

02

### Understand

Ask questions grounded in the selected study material.

03

### Practice

Generate flashcards and quizzes with broader, less repetitive coverage.

04

### Continue

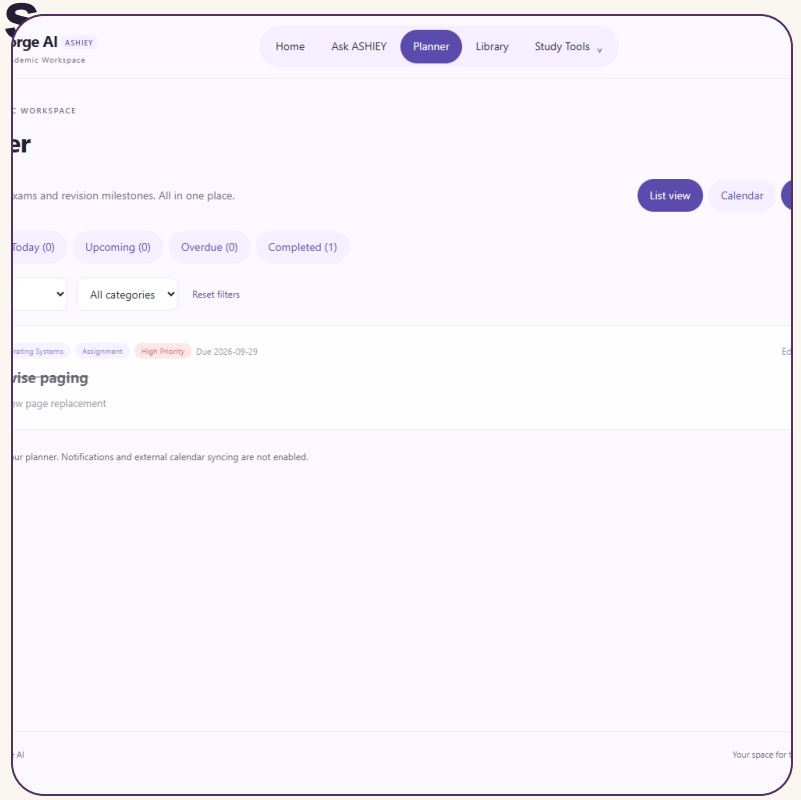
Use a dashboard, planner, streaks and profile preferences to stay consistent.

**The product goal is not just AI answers. It is a repeatable study workflow.**

WHAT STUDENTS CAN DO

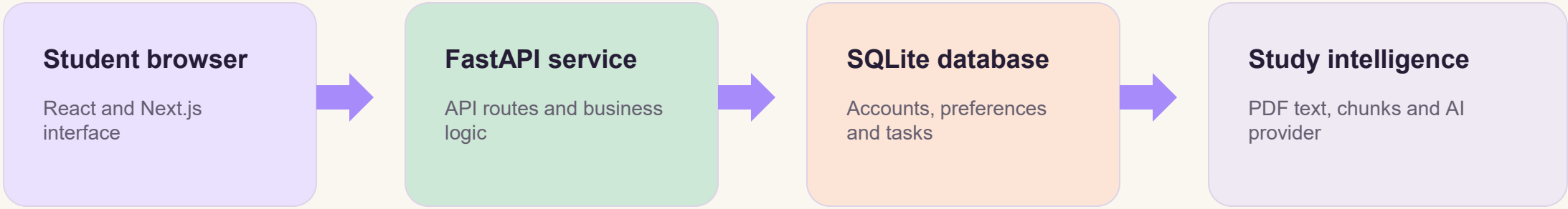
# One application, several focused study actions

| | |
|---|---|
| **Material library** | Store uploaded PDFs and choose the active study source. |
| **Study chat** | Ask a topic-specific question and receive a grounded response. |
| **Flashcards** | Create short review prompts for recall practice. |
| **Quizzes** | Attempt varied MCQs and receive immediate feedback. |
| **Planner** | Organise tasks and deadlines around a study routine. |



SYSTEM ARCHITECTURE

# A React frontend communicates with a FastAPI backend and local services.

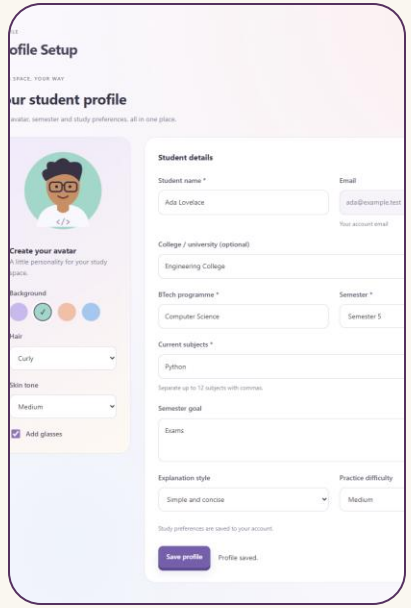



**The browser never directly handles database queries or provider calls. The backend manages these steps.**

BACKEND WORKFLOW

# A request travels through clear API steps before the student sees a result.

| | |
|---|---|
| **Authentication** | Create account, sign in and protect user-specific actions. |
| **Material processing** | Read PDF text, split it into searchable chunks and save useful metadata. |
| **Study endpoints** | Receive questions, flashcard requests and quiz requests from the frontend. |
| **Persistence** | Store profiles, tasks, files and selected study context in SQLite. |



GROUNDED AI FLOW

# The selected PDF supplies context for answers and practice material.

1

## Upload PDF

PyMuPDF extracts text from the student's document.

2

## Chunk content

The backend divides text into smaller searchable pieces.

3

## Find context

TF-IDF retrieval selects relevant chunks for a question.

4

## Generate response

Gemini is used when available, with local handling for reliability.

**Why this matters: the system can answer using material the student selected, instead of only a generic prompt.**

FRONTEND ARCHITECTURE

# The frontend makes study tasks clear.

Frontend section presented by Manik Jasrotia.

## App pages

Dashboard, library, chat, quizzes, planner and profile pages.

## Reusable components

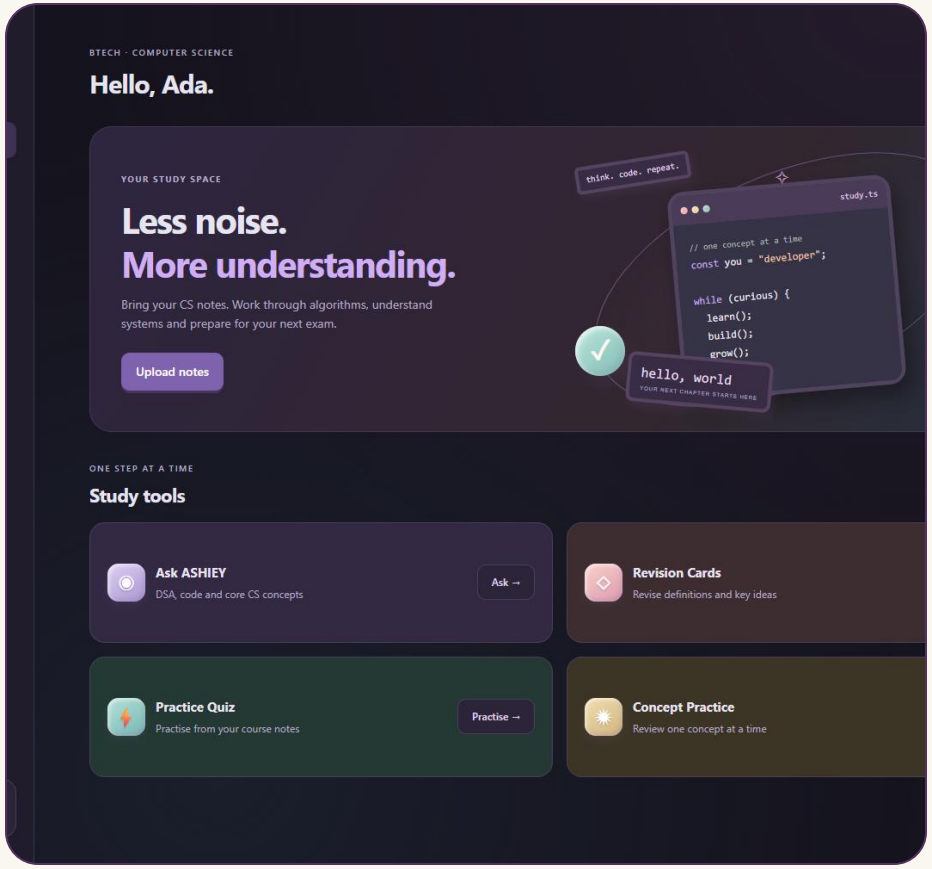
Navigation, cards, controls and modal patterns are shared across screens.

## State and APIs

The frontend calls backend routes and refreshes visible study data.

## Responsive styling

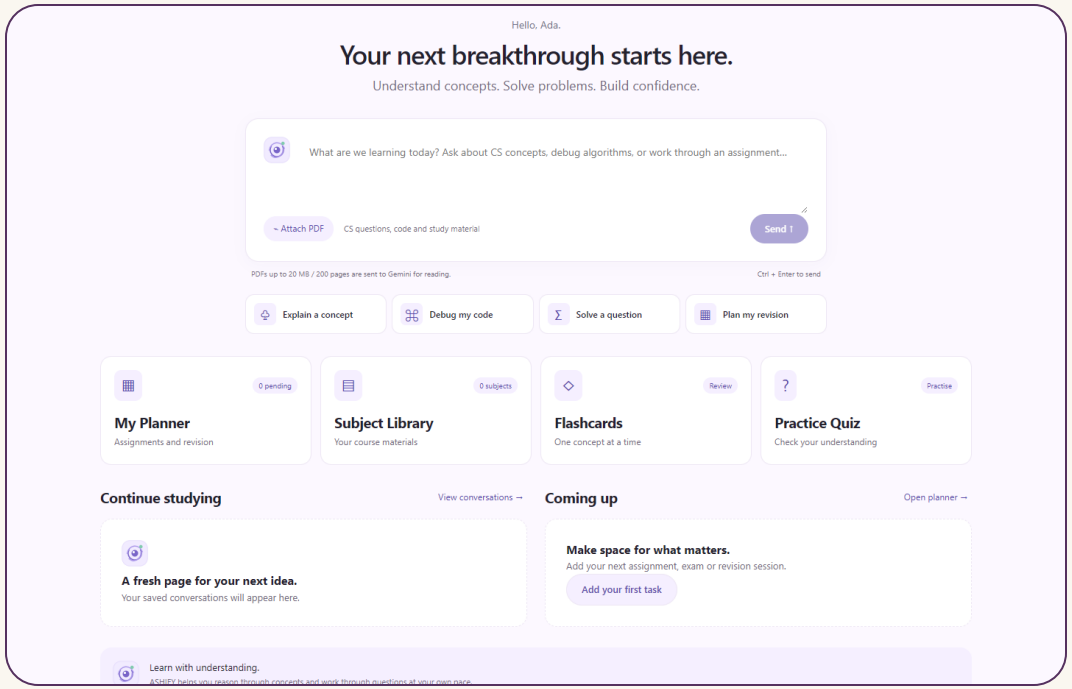
Layouts adapt to common desktop and mobile widths.



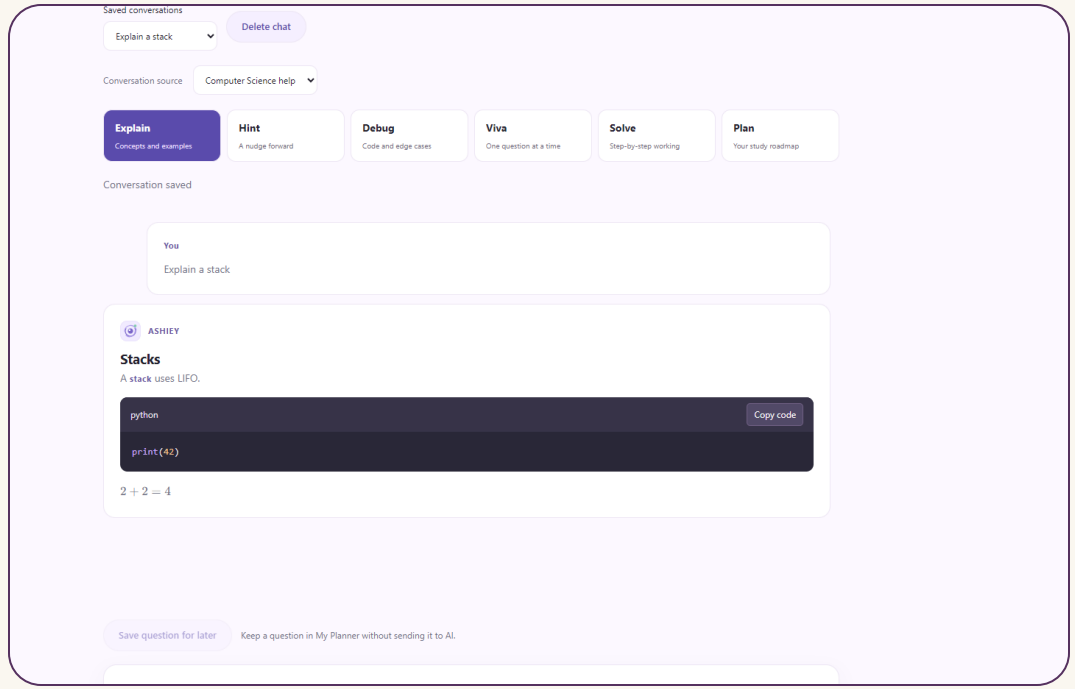
DESIGNED FOR A HUMAN STUDY ROUTINE

# Warm surfaces, clear hierarchy and fewer distractions support regular use.

The interface puts the next study action in view instead of making students search through technical menus.




DASHBOARD

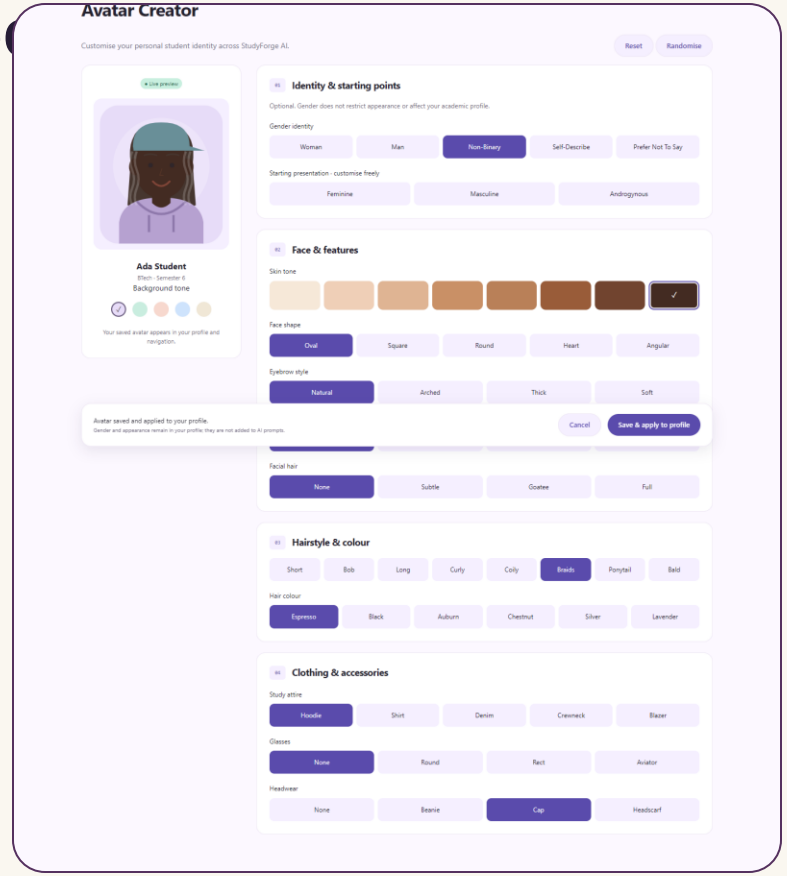



CONTEXTUAL STUDY CHAT

# Personalisation makes the product feel owned by the student.

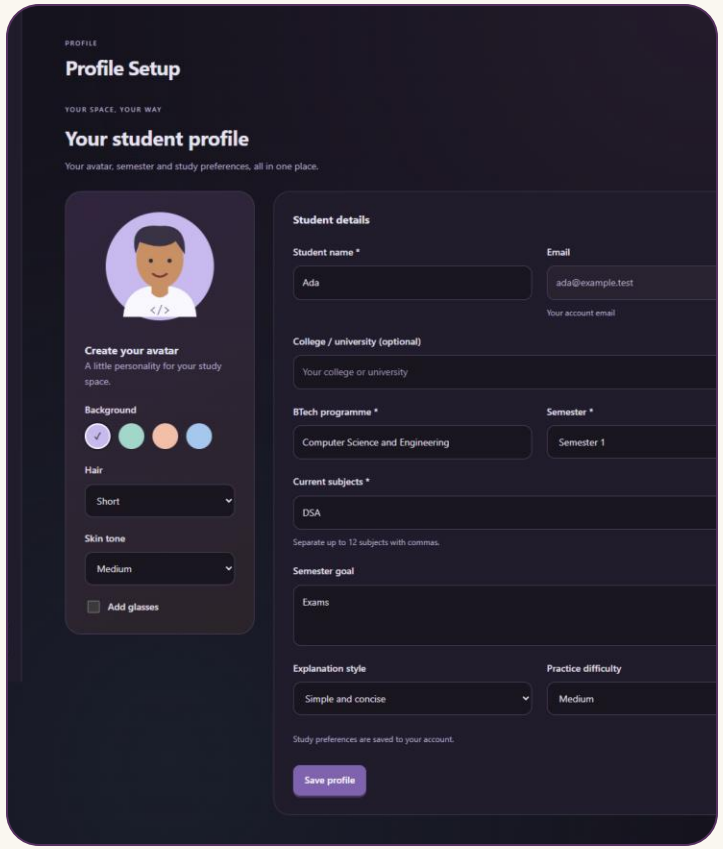
- A profile setup flow stores preferences through the backend profile API.
- Gender-aware options and custom choices are presented in a 3D-styled visual avatar experience.
- Theme controls support light and dark study environments while maintaining readable content.

FRONTEND-ONLY PRESENTATION SECTION



PRACTICE THAT STAYS USEFUL

# Flashcards and quizzes are designed to cover different ideas from the same material.




**A**

## Broader coverage

Prompts ask for several different concepts.

**B**

## Duplicate filter

Similarity checks remove near-repeat questions.

**C**

## Topic labels

Focus labels show what each item tests.

**If the external AI provider is temporarily busy, the backend creates a local source-grounded quiz from saved material chunks.**

VALIDATION AND CURRENT STATUS

# The project was checked from backend behaviour through to the visible frontend build.

✓

### Backend reliability tests

Targeted tests covered AI failure handling and workspace behaviour.

✓

### Local quiz fallback

Direct testing generated varied questions from stored study chunks.

✓

### Frontend production build

The Next.js frontend completed a production build successfully.

✓

### Browser review

Pages and study actions were checked through the running local application.

WHAT COMES NEXT

# The project has a solid student workflow and clear paths for future improvement.

## Smarter content review

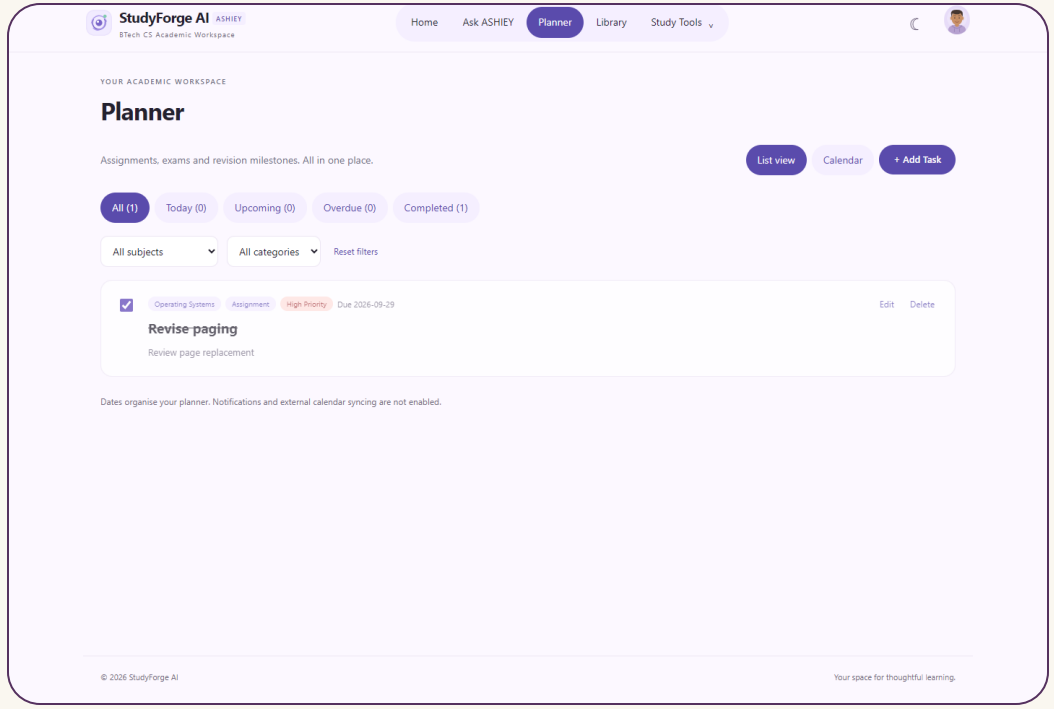
Allow students to rate cards and quizzes to improve future generation.

## Analytics

Show weak topics, time spent and revision history.

## More formats

Support notes, images and additional subject-specific templates.



## Live demo flow

1. Sign in → 2. Select study material → 3. Ask a question → 4. Generate a quiz

# Thank you

Questions and discussion

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StudyForge AI — a calmer study workspace for B.Tech students